



Fine-Tuning and Reinforcement Learning with Human Feedback

CSIT205 Generative AI – 2025



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*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*

Dr Andrew Yoong, SIM



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, with the 'U' at the top, 'O' in the middle, and 'W' at the bottom.

Everyone is welcome here.

UOW takes pride in
the diversity of our
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Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

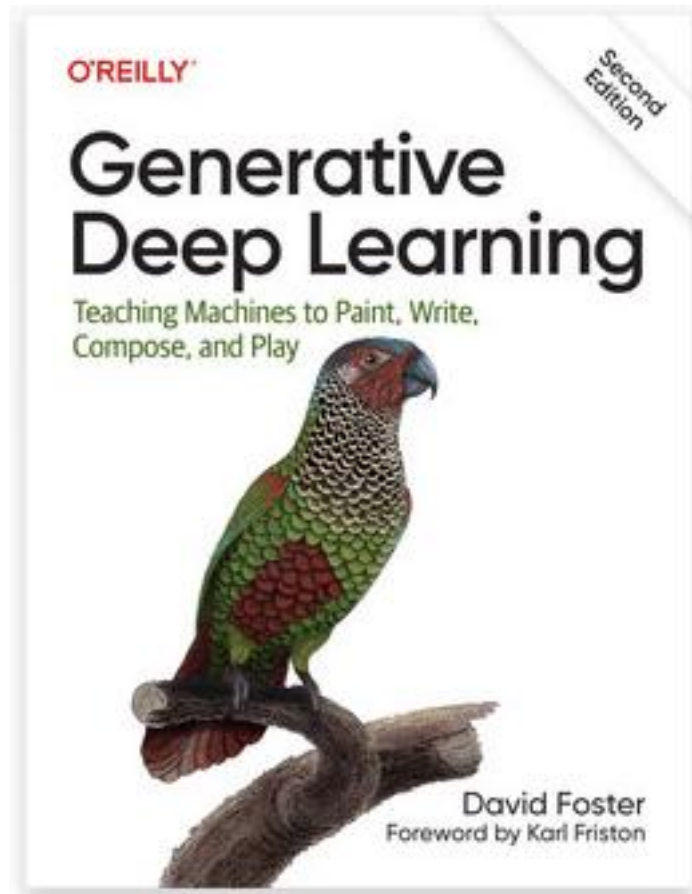
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don't forget 

This is the book we're using



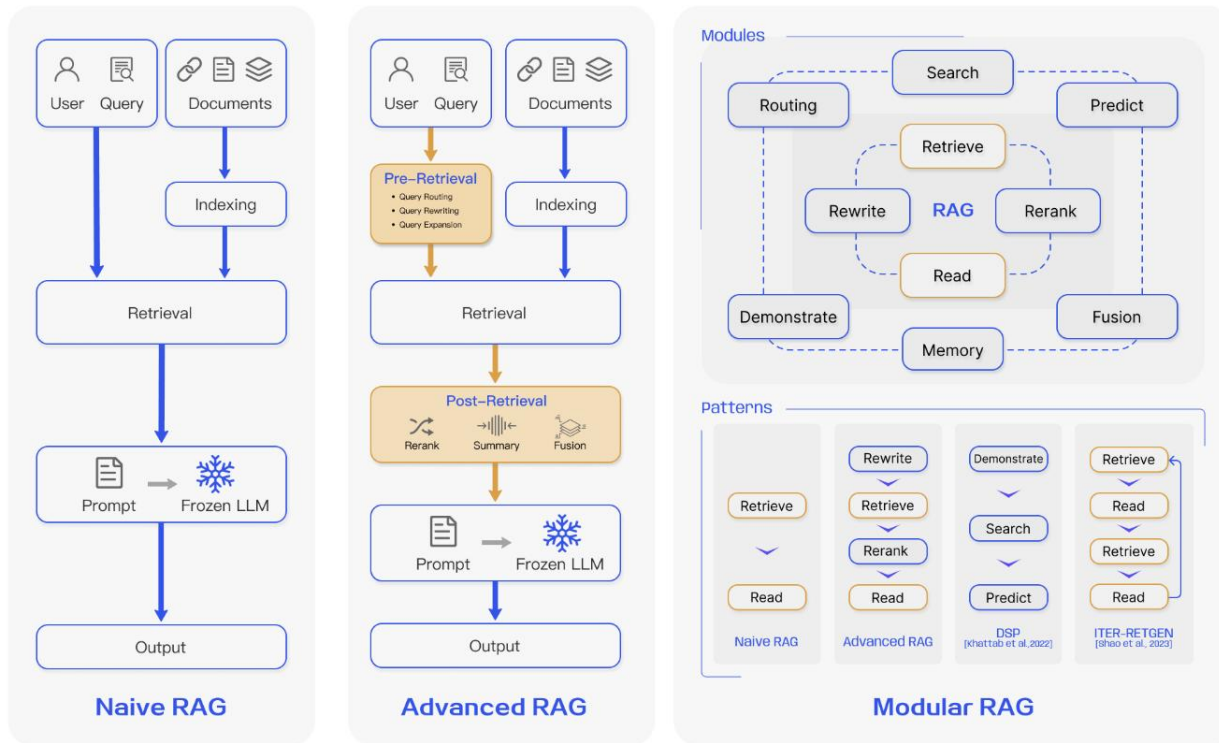
<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

What did we discuss last week?



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Types of RAG applications (Gao *et al.*, 2023)

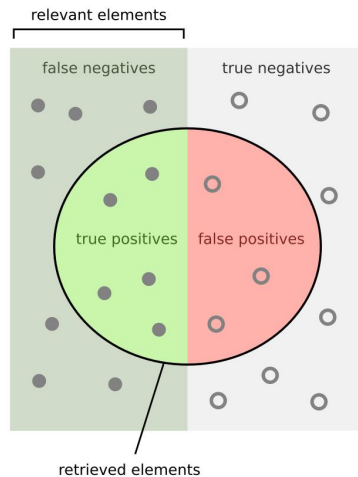


What is RAG ?
Advanced RAG
Modular RAG

Evaluation of RAGs

Caters for new knowledge resources, LLMs, pre- and post-retrieval modules

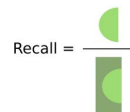
- BLEU
- ROUGE
- RaLLE (Hoshi et al., 2023):
 - R-precision: r/R where R = total number of relevant items; r = number of relevant items retrieved as top- R ;
 - Recall@5: number of items correctly recalled from a prescribed list of 5.
 - Experiments are tracked by [MLFlow](#)



How many retrieved items are relevant?

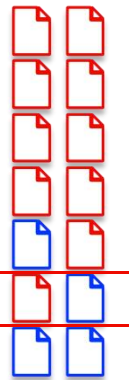


How many relevant items are retrieved?



$R\text{-precision} = 4/5$

$\text{Recall@5} = 5/5$



Recall becomes useless when lots of documents can be retrieved

Why are we here?

CSIT205 Generative AI - Module 2

Fine-tuning and RLHF

- Fine-tuning
- Reinforcement Learning with Human Feedback (RLHF)



Fine-tuning



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Your model is not generating the right output...

And you've tried:

- *Prompt engineering to be more specific about the task and output you require.*
- *RAG to provide additional context.*

But it still doesn't work. What's can we do to further enhance performance?

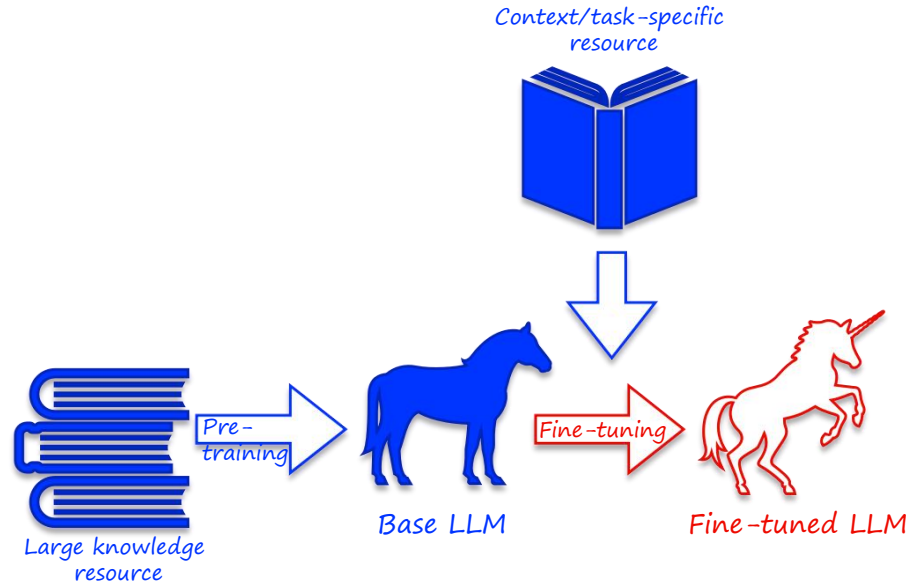
Fine-tuning!



Image of worried data scientist, generated by MS Copilot

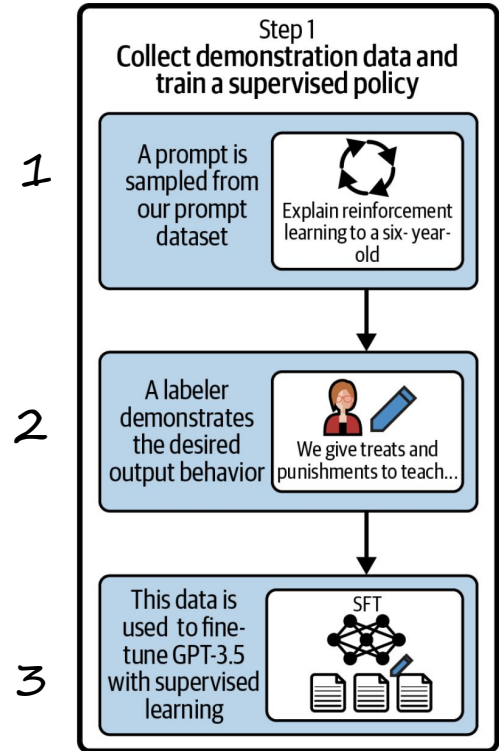
Fine-tuning in a nutshell

- Tweak your model to learn from examples specific to your context.
- It lets you adjust parameters, not the model architecture.
- For LLMs, it allows you to rely on general language skills learned by the model, and adapt it to suit your particular task or domain.
- Usually comes after prompt engineering and RAG, because it is more resource intensive.



The supervised fine-tuning (SFT) process

1. *Select a prompt / task that you'd like to optimise for.*
2. *Provide the desired output to provide the training process with examples of good output.*
3. *Retrain the model to update its parameters.*



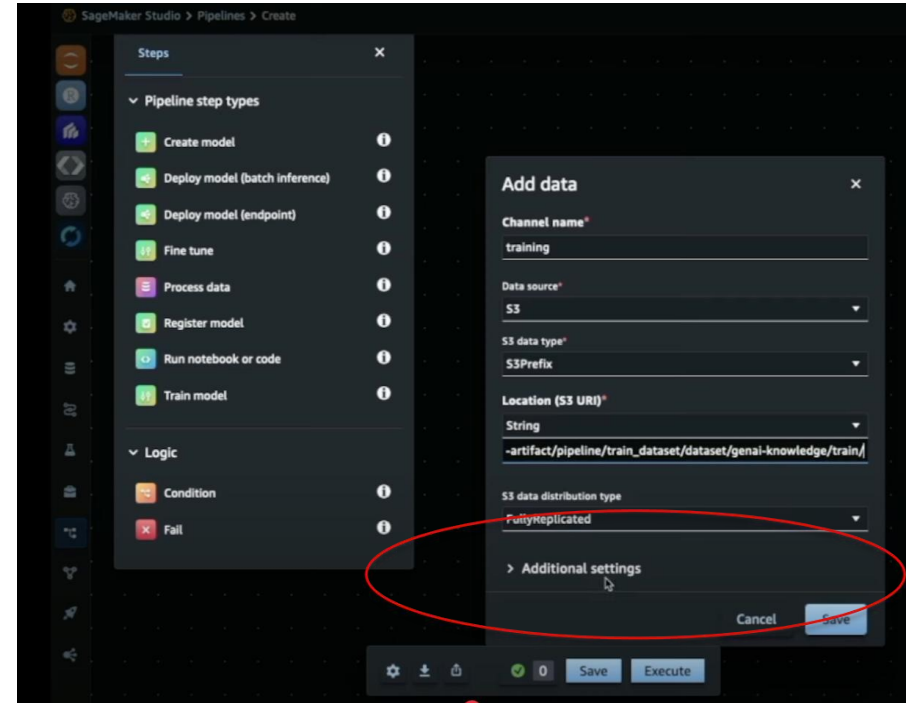
Foster, 2023

How many samples do you generally need?

- Depends, but 500-1000 is common for a single task

Popular tools for SFT

- AWS SageMaker (see image):
SageMaker Notebooks, Pipelines, or [Jumpstart](#)
- Google Colab
- [Google Vertex AI](#)
- [Unsloth](#): works locally and speeds up fine-tuning
- [MLX](#): Apple Silicon only



<https://youtu.be/LV-Vu2WRdVA?si=qxdQO7LkAnC6I>

Challenges for supervised fine-tuning

- ✗ *Training all parameters is time-consuming and expensive*
- ✗ *Collecting labelled samples is labour-intensive*
- ✗ *A tendency to prioritise recent data over older data leads to catastrophic forgetting of older skills.*
- ✗ *A tendency to prioritise recent data can lead to overfitting towards newer skills.*

Parameter-efficient Fine-tuning (PEFT)



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Parameter Efficient Fine-Tuning (PEFT)

- Collection of measures to speed up training:

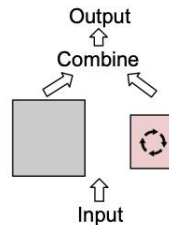
- Additive Fine-tuning
- Selective Fine-tuning
- Reparameterised Fine-tuning
- Hybrid Fine-tuning

- Key mechanism: parts of the model remain 'frozen' (grey), and only a small portion of parameters is updated (red)

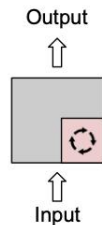
→ Faster training

→ Prevents catastrophic forgetting

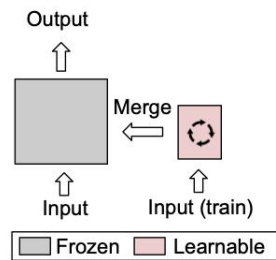
Additive PEFT



Selective PEFT



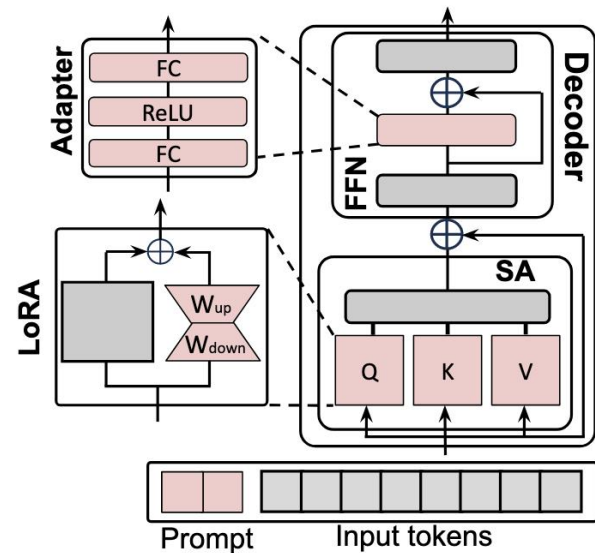
Reparameterised PEFT



Han et al., 2024

Three common patterns of PEFT

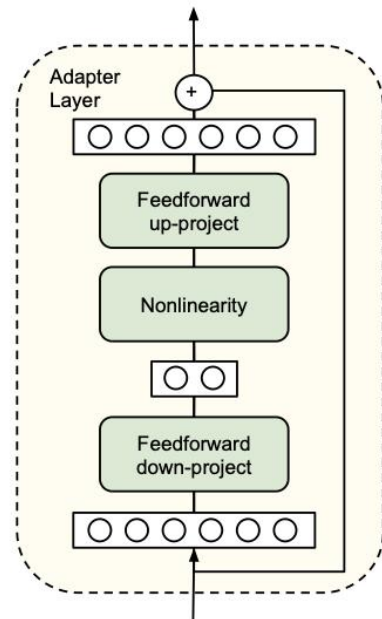
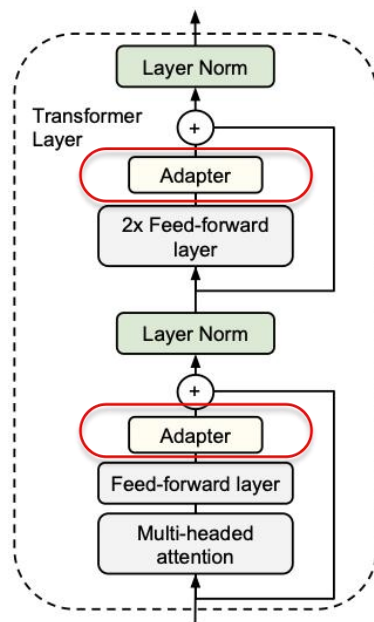
- *Adapter (additive PEFT)*
- *Low Rank Adaptation (LoRA)*
(reparameterised PEFT)
- *Soft prompts (additive PEFT)*



Han et al., 2024

Adapter

- Added to each transformer layer
- Pre-trained model weights remain frozen
- ✓ Only adapter weights are retrained / fine-tuned on the new task
- ✗ usually runs sequentially, which adds to inference latency



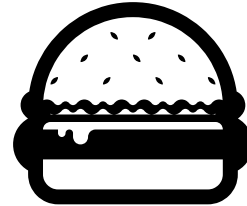
Houlsby et al., 2019

10-minute Break!

This will help you learn



10 minutes





Which pattern is not common for PEFT?

A. Additive PEFT

B. Low Rank Adaptation (LoRA)

C. Hard Prompts

D. Soft Prompts



True or False

Catastrophic forgetting is a common problem in supervised fine-tuning that can be overcome with PEFT Adapters.

True

Low Rank Adaptation (LoRA)



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Low Rank Adaptation (LoRA) (Hu *et al.*, 2021)

- Motivation: there exists an 'intrinsic lower dimension', rank r (Aghajanyan *et al.*, 2020) that is just as effective as retraining all parameters for a given task.

- 'Bottleneck adapter', because decomposes the matrix while maintaining as much information as possible.

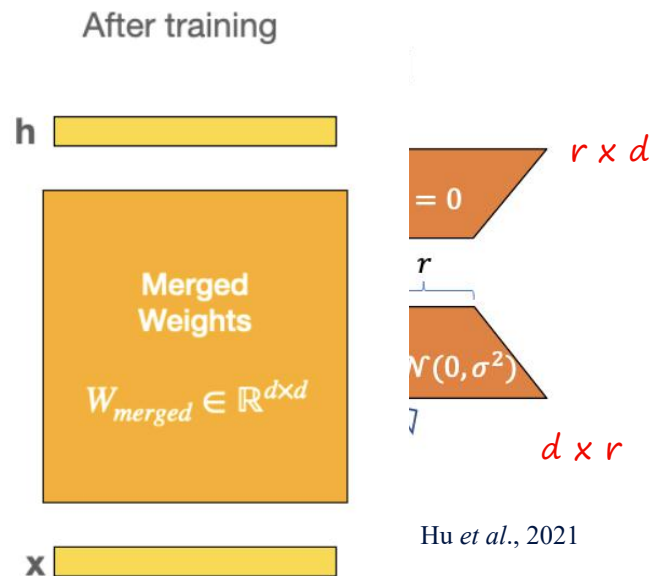
→ All pre-trained model parameters are frozen!

- Training Process:

- Determine a rank r (usually small, like 1 or 2)
- Learn matrices A ($d \times r$) and B ($r \times d$)

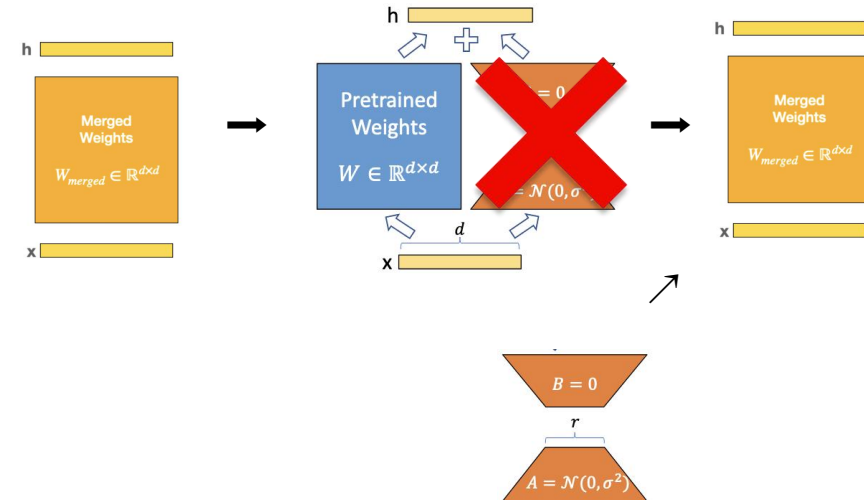
- Merge $A * B$ with pre-trained weights

If $d=64$, and $r=2$, then n learned parameters is $64 * 2 + 2 * 64 = 256$, instead of $64 * 64 = 4096$



Advantages of LoRA (Hu *et al.*, 2021)

- ✓ build many LoRA modules for different tasks → save the LoRA weights A and B, un-merge and merge with a different LoRA module.
- ✓ training is more efficient, because $r*d + d*r < d*d$
- ✓ No inference latency compared to a fully retrained model and faster than adapters, because it weights are merged.
- ✓ can be used with other methods, like prompt-tuning.



LoRA Code example

Import the LM as `model` (not shown)

Set LoRA parameters. Here the rank r is set to 2.

target modules are the pre-trained weights Q and V to be reparameterised.

Get the new LoRA model

The LoRA trainable parameters are 0.02% of the pre-trained model's parameters!

```
from peft import LoraConfig, get_peft_model
```

```
config = LoraConfig(  
    r=2,  
    lora_alpha=16,  
    target_modules=["q_proj", "v_proj"],  
    lora_dropout=0.1,
```

such parameter

```
classifier"],
```

```
model(model, config)
```

```
lora_model)
```

```
all params: 1236027392 || trainable%: 0.02
```

much wow

many efficient

Note that you can also use Unsloth to speed up things

https://huggingface.co/docs/peft/main/en/developer_guides/lora

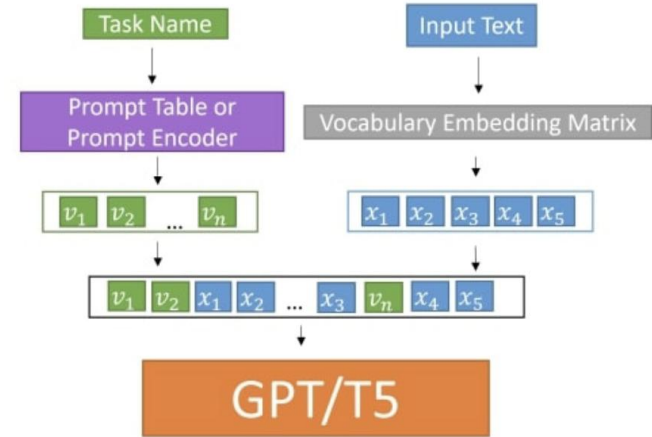
Soft Prompts



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Soft-prompts (Lester *et al.*, 2021)

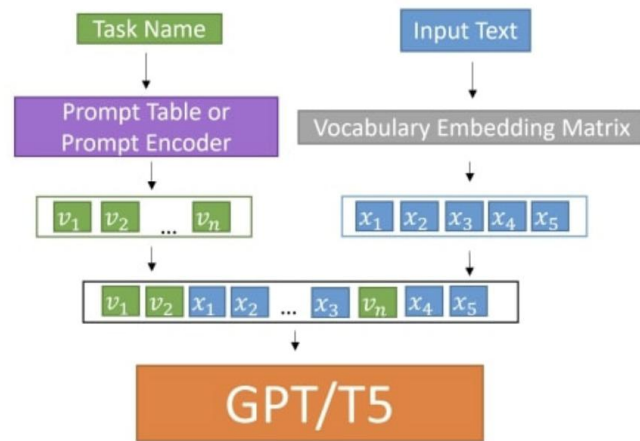
- Added before frozen pre-trained model to provide more information on task at hand → 1 model, multiple tasks.
- Learns a prompt (read: trains a model) with multiple tokens $\{p_1, p_2, \dots, p_n\}$ for each task. This is flexible as opposed to fixed, 'hard' prompts, hence 'soft-prompts'
- Training process:
 1. Initialise prompts for tasks with actual words (e.g. 'technology',) from the vocabulary (i.e. discrete 'embedding' tokens)
 2. Learn prompts of continuous embedding tokens
 3. Result: 'Prompt Table' or 'Prompt Encoder'



Shah, 2024

Pros and cons of Soft-prompts (Lester *et al.*, 2021)

- ✓ Only prompts are trained / fine-tuned on new tasks
- ✓ Faster fine-tuning
- ✗ Runs sequentially, which adds to inference latency
- ✗ Continuous embedding tokens (e.g. a token close to 'technology' is learned) make interpretability harder



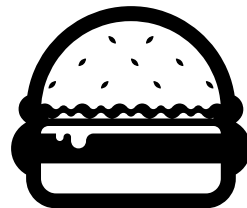
Shah, 2024

5-minute Break!

This will help you learn



5 minutes



Which problems does LoRA address? Select all that apply.



- A. More efficient training due to a reduction of learnable parameters.
- B. The labour-intensity of collecting labelled examples.
- C. Increased latency for other PEFT methods such as Adapters and Soft Prompts.
- D. Catastrophic forgetting



Fill in the blanks

Soft Prompts run in _____, which adds to _____ latency

A. parallel, training

B. parallel, inference

C. sequence, training

D. sequence, inference

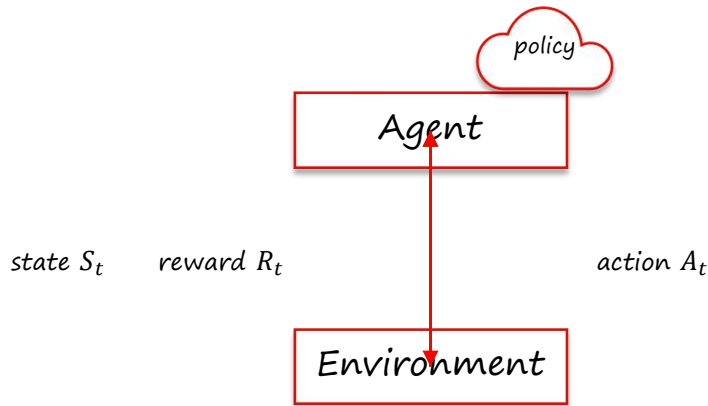
Reinforcement Learning with Human Feedback (RLHF)



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Reinforcement Learning (RL)

- The Agent learns from the feedback on past actions.
- Agent adopts a policy to determine which action to take.
- The environment returns feedback to the Agent in the form of a new environment state and a reward



Challenges for RL in the context of LLMs

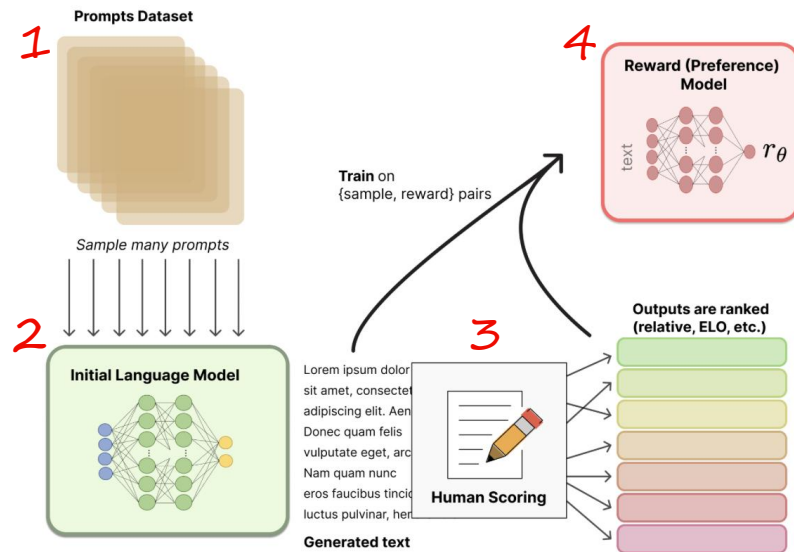
- ✗ *LLMs are non-deterministic (read; creative)*
- ✗ *Quality of an LLM's response is often subjective to individuals*
- ✗ *Multiple answers may be correct*

```
>>> when was Taylor Swift born?  
Taylor Alison Swift was born on December 13, 1989, in Reading,  
Pennsylvania, United States.  
  
>>> when was Taylor Swift born?  
I can provide you with the correct information! According to my  
knowledge, Taylor Swift was born on:  
  
December 13, 1989
```

Generated with LLaMA3.2 on Ollama

Reinforcement Learning with Human Feedback (RLHF)

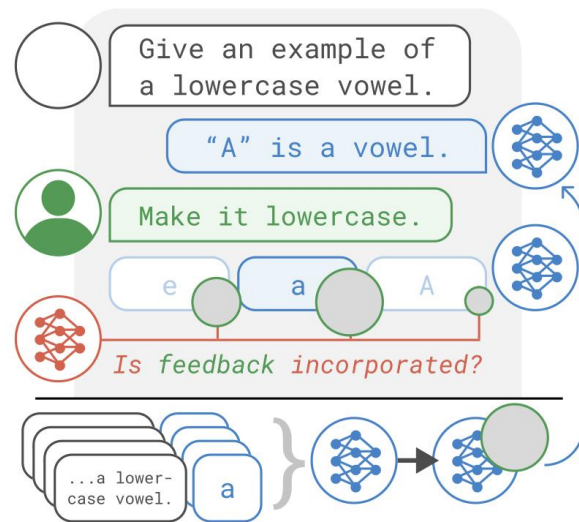
- Quality of LLMs is subject to human preferences → human feedback determines the reward
- Process:
 1. Collect and present prompts.
 2. Collect multiple LLM responses per prompt.
 3. Human provides a ranking (reward).
 4. A reward model (Stiennon et al, 2022) is trained to calculate the reward for each LLM response.
 5. A policy (not shown) is used to train the initial LM using RL with the reward model providing feedback.



Source: <https://huggingface.co/blog/rlhf>

RLHF Methods

- *Preference-based: a human picks the best response from a set of responses.*
- *Rating-based: a human rates the responses.*
- *Language-based: feedback in natural language, e.g. 'change the name*
- *Miscellaneous: e.g. a model generates potential feedback to guide humans.*



Language-based feedback (Scheurer et al., 2024)

Alignment of AI assistants (a.k.a. HHH; Askell *et al.*, 2021)

The following criteria help with providing feedback:

- *Helpfulness: the assistant will always try to do what is in the humans' best interests*
- *Honesty: the assistant will always try to convey accurate information to the humans and will always try to avoid deceiving them*
- *Harmlessness: the assistant will always try to avoid doing anything that harms the humans*



image of a superhero with HHH on his chest, MS Copilot

Advantages of RLHF

- ✓ Improved model performance due to additional training
- ✓ Better alignment due to human feedback
- ✓ More robust to changes in tasks due to variety of prompts and human evaluations
- ✓ Better safety and security because of the human-in-the-loop



image of a successful woman holding a document that says RLHF, comic style, MS Copilot

Disadvantages of RLHF

- ✗ Time-consuming to collect human feedback
- ✗ Labour-intensive to collect human feedback
- ✗ Humans may be biased
- ✗ Humans may lack domain knowledge



image of a defeated superhero that wears HHH on his chest, MS Copilot

Wrap-up



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Summary

- Supervised Fine-tuning is the next step after prompt engineering and Retrieval Augmented Generation (RAG)
- We have four types of Parameter-Efficient Fine-Tuning (PEFT): additive, selective, reparameterised, and hybrid
- We see three common patterns of PEFT: Adapters, Low Rank Adaptation (LoRA), and Soft Prompts
- Reinforcement Learning with Human Feedback (RLHF) allows LLMs to learn from human feedback

For next lecture

Emerging Trends in Enhancing Model Performance

Take Home Message...

*Consider the message that you want to create when
designing your interfaces.*



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Things to think about

What is the purpose of prototyping?

*What is the difference between Lo-fi
and Hi-fi designs?*

*What is a storyboard? When can
these be used?*

What should you show your client?

References and Resources

- Aghajanyan, A., Zettlemoyer, L., & Gupta, S. (2020). *Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning* (No. arXiv:2012.13255). arXiv. <https://doi.org/10.48550/arXiv.2012.13255>
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Questions



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